# ⛽ Biofuel & Energy: Reading · ANSWER KEY

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

Name: _____

## From Algae Oil to Fuel

Matter is anything that has mass and takes up space, like oil, water and air. Energy has no mass; it is the ability to make things move or change. Potential energy is stored energy, like water behind a dam. Kinetic energy is the energy of motion, like water rushing out of the dam.

Energy can be transferred, which means it moves to a new place (a hot pan warms your hand). It can also be transformed, which means it changes form (fuel burns and becomes heat). Energy is never made or destroyed, but some always turns into wasted heat.

Fats and oils are lipids. A fat is made of glycerol plus three fatty acids. Lipids store about 2.5 times more energy than sugar, gram for gram. Algae change sugar into oil to store energy, and that oil can be made into biodiesel.

To make biodiesel, oil is mixed with methanol and a catalyst, which makes biodiesel and glycerol. Methanol and the catalyst are dangerous, so only the teacher handles them. When fuel burns (combustion), fuel + oxygen → carbon dioxide + water + energy. In a bus, chemical energy becomes heat and then mechanical energy that moves the bus.

Some reactions are exothermic: they release heat and the temperature goes up. Others are endothermic: they take in heat and the temperature goes down, like baking soda and vinegar. Biodiesel is renewable because algae and plants grow back fast. Fossil fuels (coal, oil, natural gas) take millions of years to form. In the carbon cycle, plants take CO2 out of the air, and burning and breathing put it back.

### Check your reading

1. What is the difference between energy transfer and energy transformation?
**Transfer: energy moves to a new place. Transformation: energy changes form.**

2. Why is biodiesel called renewable?
**It is made from plants/algae, which grow back in a short time.**

3. Baking soda and vinegar got colder. Which kind of reaction is this?
**Endothermic.**